AAT · MANAGEMENT AND ENTREPRENEURSHIP (23CS7PCMNE) · DEPT. OF CSE, BMSCE

SkyPulse Presenter's Guide

Drone-Based Emergency Medicine Delivery

Detailed business, production, operations, marketing, organizational and financial plans, risk assessment, a presentation script for each slide, and viva questions.

Slides	Topic	Presenter	Time
1–3	Title, Introduction, How it works	Member 1	2 min
4–5	Market & competition, Production plan	Member 2	2 min
6–7	Operations plan, Marketing plan	Member 3	2 min
8–10	Organizational plan, Risk, Financial plan	Member 4	3.5 min
11	Conclusion & questions	Member 1	0.5 min
	Total		10 min

Replace "Member 1–4" with your team's names. Figures match the report, PPT and Excel model.

1 Business plan summary

Slides 2–4 · Member 1 (Introduction, how it works) and Member 2 (market and competition)

1.1 The problem

- In a medical emergency the first hour, the **golden hour**, decides survival. In rural Karnataka the medicine is often not in the building: most PHCs have no blood bank, and one snakebite can need up to 10 vials of anti-snake venom (ASV).
- The nearest stock is typically 40–120 km away by road, which takes **1 to 4 hours**.
- India loses about **50,000 people a year to snakebite**; Karnataka recorded 13,235 cases and 100 deaths in 2024.
- A CAG audit found Karnataka's 108 ambulances met the 30-minute target in only **72% of cases**, with delays of 56–80 minutes in individual cases.

1.2 The solution: SkyPulse

SkyPulse Aero Health Pvt. Ltd. (proposed) is a **drone-as-a-service medical logistics company**. It flies blood units, ASV, life-saving drugs, vaccines and diagnostic samples between district hospitals, CHCs, PHCs and private hospitals in **15–30 minutes**, using temperature-controlled VTOL drones.

Step	What happens
1. Order	A doctor or nurse at a registered facility orders through the app, WhatsApp or toll-free IVR
2. Pack	The hub pharmacist verifies the requisition and packs a sealed, temperature-logged (2–8 °C) pod
3. Fly	A VTOL drone (5 kg payload, 80–100 km/h) flies a DGCA-cleared route
4. Deliver	Parachute drop or landing at the facility's drop zone; OTP handover; temperature log uploaded

Hub-and-spoke model: one hub at each district hospital (next to its blood bank and pharmacy) serves every facility within a 40 km radius.

1.3 Evidence that it works

- **Rwanda (Zipline):** median blood delivery time fell to 41 minutes against 2+ hours by road, and blood wastage fell by about two-thirds (Lancet Global Health, 2022).
- **ICMR i-Drone (India):** vaccine delivery in Manipur and Nagaland, blood in Delhi NCR, and tissue samples from Karkala to Manipal in Karnataka (37 km in 15–20 min vs 50–60 min by road).
- **Meghalaya study (Gupta et al., 2025):** drone round trip 50 min vs 240 min by road.

1.4 Market and competition

- India drone market: **USD 0.47 B (2025) → USD 1.39 B (2030)**, CAGR 24.4%. Delivery drones are the fastest-growing segment (CAGR 50.4% to 2030).
- Karnataka has **2,138 PHCs across 31 districts**, each a potential delivery point.
- Policy support: Drone Rules 2021 (about 90% of airspace is a green zone), ₹120 Cr PLI scheme, Drone Shakti mission, ban on imported drones (Make in India), and a BVLOS framework expected by end-2026.

- Market sizing (Karnataka): TAM about ₹160 Cr/yr, SAM about ₹80 Cr/yr, SOM ₹29.8 Cr by Year 5 (about 37% of SAM).

Competitor	Focus	Gap SkyPulse fills
Airbound (Bengaluru)	Urban lab samples for Narayana Health; raised US$37M	No rural emergency / ASV / blood to PHCs
Skye Air Mobility	Parcel and health logistics	No cold-chain emergency SLA for government
TSAW Drones, Redwing Labs	Pilots; North-East terrain	Not scaled in Karnataka
Road / 108 ambulance (substitute)	Moves patients and goods by road	1–4 hours, ₹2,902 per round trip

1.5 Business model (revenue streams)

Stream	Price (Year 1)	Customer
Emergency delivery	₹1,400 per delivery	Hospitals / PHCs
Scheduled delivery (samples, resupply)	₹500 per delivery	Hospitals, labs
Hospital subscription (priority SLA + dashboard)	₹25,000 per month	Private hospitals
District contract (routine PHC resupply)	₹36 lakh per year	NHM / State Health Dept.

SCRIPT **Member 1 · Slide 2 · about 1 minute**

Good morning. We are presenting SkyPulse, a drone-based emergency medicine delivery business for Karnataka.

Imagine a woman bleeding after childbirth in a primary health centre in rural Kolar. The doctor is there, but the blood is not. The nearest blood bank is 50 kilometres away, and by road that is over an hour each way. In an emergency, that hour decides whether she lives.

This is not rare. India loses around fifty thousand people a year to snakebite alone, and Karnataka recorded over thirteen thousand snakebite cases in 2024. A CAG audit found that our 108 ambulances meet the 30-minute target in only 72 percent of cases.

Our answer is SkyPulse: a network of temperature-controlled drones that delivers blood, anti-venom and life-saving drugs to any health centre in 15 to 30 minutes.

SCRIPT **Member 1 · Slide 3 · about 1 minute**

Here is how it works, in four steps. A doctor at a registered health centre places an order through our app, WhatsApp or a toll-free number. At the district hospital hub, the pharmacist verifies the request and packs the medicine in a sealed, temperature-logged pod. A vertical take-off drone flies it at 80 to 100 kilometres per hour. It is dropped at the health centre, and the staff confirm receipt with an OTP.

We run a hub-and-spoke model. Each hub sits at a district hospital and serves every facility within 40 kilometres.

We earn money in three ways: a fee per delivery, ₹1,400 for emergencies and ₹500 for scheduled runs; a monthly subscription of ₹25,000 for private hospitals that want a guaranteed service; and annual contracts of ₹36 lakh per district with the State Health Department.

SCRIPT **Member 2 · Slide 4 · about 1 minute**

Is there a market? India's drone market is expected to triple from 0.47 to 1.39 billion dollars by 2030, and delivery drones are the fastest-growing part, at 50 percent a year. Karnataka alone has over 2,100 primary health centres, and every one is a delivery point.

The timing is right. The Drone Rules 2021 opened 90 percent of Indian airspace, the government is funding drone manufacturing through the PLI scheme, and a framework for long-range flights is expected by the end of this year.

Competitors like Airbound and Skye Air focus on urban lab samples and parcels. Nobody runs a rural emergency network in Karnataka. That is our gap.

Our edge is simple: we are faster than road, cheaper than road, and every delivery carries a verified cold-chain record.

If asked: Why not just stock blood and ASV at every PHC? Blood expires in 35–42 days and needs a licensed blood bank; small PHCs use too little to keep stock fresh, so it gets wasted. Rwanda's drone network cut blood wastage by about two-thirds by keeping stock central and flying it on demand.

2 Production plan

Slide 5 · Member 2

SkyPulse is a **service business**. Its "production" is the readiness of the drone fleet, the hubs and the software that together generate deliveries.

2.1 Make-or-buy decisions

Component	Make / Buy	Reason
VTOL drone airframe and avionics	Buy	From DGCA type-certified Indian manufacturers; foreign drone imports banned since 2022
Cold-chain payload pod + IoT temperature logger	Make (in-house assembly)	Our differentiator; insulated 5 kg pod, gel packs, tamper seal, parachute release
SkyPulse Command software	Make	Ordering app, dispatch engine, route optimiser, hospital dashboard; core IP of a CSE team
Hub civil works (pads, shelter)	Subcontract	Local contractors
Routine maintenance (A-checks)	Make	In-house technicians at each hub
Overhaul (C-checks)	Subcontract	Manufacturer's annual maintenance contract

2.2 Drone specification

Parameter	Specification
Type	Hybrid VTOL fixed-wing (takes off like a helicopter, cruises like a plane)
Payload	Up to 5 kg (about 10 blood units or 50 ASV vials)
Range / radius	100 km round trip; 40 km operating radius
Speed	80–100 km/h
Safety	Geofencing, return-to-home, emergency parachute, redundant GPS
Compliance	DGCA type certificate, UIN, NPNT (No Permission No Take-off), Remote ID
Cost	₹14 lakh per drone (Indian listings ₹3.6–18 lakh)

2.3 Physical plant (each hub)

- Located on a district hospital campus next to the blood bank and pharmacy, under an MoU with the State Health Department.
- Two 6 m × 6 m landing pads with wind socks and fencing.
- Charging room with fire-rated storage for 30 battery packs.
- Packing bay with a pharmacy-grade refrigerator (2–8 °C) and freezer for gel packs.
- Ground control station (GCS) room for remote pilots, with backup internet.
- Each receiving PHC gets a painted 3 m × 3 m QR-coded drop zone (about ₹15,000 per site).
- Staffing per hub: 6 drones, 3 remote pilots, 2 technicians, 1 hub manager.

2.4 Equipment and capital cost (Year 1)

Item	Qty	Unit cost (₹ lakh)	Total (₹ lakh)
VTOL drone with cold-chain pod	12	14.0	168.0
Hub setup (pads, chargers, cold storage, GCS)	2	22.0	44.0
Technology platform development	1	60.0	60.0
Total Year-1 capex			**272.0**

2.5 Suppliers

Input	Supplier type
Drones and spares	DGCA type-certified Indian drone manufacturers (PLI-supported)
Battery packs	Indian Li-ion pack assemblers
Cold-chain boxes, gel packs	Medical cold-chain packaging firms
IoT loggers, 4G SIMs	Indian IoT vendors, telecom operators
Cloud hosting	Indian data-centre region (data stays in India under the DPDP Act 2023)
Insurance	General insurers (third-party insurance is mandatory under Drone Rules 2021)

2.6 Capacity

A 45-minute round trip allows up to 12 flights per drone in a 10-hour shift. We plan conservatively for 5 deliveries per drone per day in Year 1, rising to 10 by Year 5. That gives **19,800 deliveries in Year 1** and **2,77,200 in Year 5**, with spare capacity kept for surges such as monsoon floods and snakebite season.

SCRIPT **Member 2 · Slide 5 · about 1 minute**

Our production plan. We are a service company, so production for us means having drones, hubs and software ready to fly.

We do not build drones. India banned foreign drone imports in 2022, so we buy certified drones from Indian manufacturers. What we build ourselves is what makes us different: the cold-chain pod with its temperature logger, and the SkyPulse software that takes orders and dispatches drones.

Each drone carries up to 5 kilograms, about 10 units of blood, flies 80 to 100 kilometres an hour and has a parachute and geofencing for safety. It costs about ₹14 lakh.

Each hub sits at a district hospital, with landing pads, a charging room, a pharmacy fridge and a control room, run by six people. In Year 1 we set up two hubs with 12 drones, a capital outlay of ₹272 lakh.

If asked: Why VTOL fixed-wing and not a quadcopter? A quadcopter hovers well but has short range. A fixed-wing flies far on little energy but needs a runway. VTOL fixed-wing takes off vertically from a hospital pad and still covers 100 km.

3 Operations plan

Slide 6 · Member 3

3.1 Description of operations

A hub at each district hospital serves every facility within 40 km, which covers 60–90 PHCs and CHCs in a typical district. Roll-out starts with **Chikkaballapur and Kolar (Year 1)** and then Tumakuru, Chitradurga, Raichur and Kodagu, choosing districts with poor road access and high snakebite or maternal-emergency rates first. The network grows from 2 hubs to 14 by Year 5.

3.2 Flow of an order (target: under 35 minutes)

1. **Order:** a doctor or nurse at a registered facility orders via the app, WhatsApp bot or toll-free IVR, choosing item, quantity and urgency.
2. **Verification (3 min):** the hub pharmacist or blood-bank officer checks the requisition and the doctor's credentials, and confirms stock.
3. **Packing:** the item goes into the sealed cold-chain pod; the temperature logger starts.
4. **Pre-flight:** the remote pilot runs the checklist; the system checks weather and airspace on DGCA Digital Sky (NPNT) and plans the route.
5. **Flight (15–25 min):** the drone takes off vertically and cruises along an approved corridor, monitored live.
6. **Delivery:** parachute drop or landing at the QR-coded drop zone; staff confirm with an OTP; the temperature log is uploaded.
7. **Return:** the drone flies back, the battery is swapped and a post-flight check is done.
8. **Billing and records:** the delivery is invoiced automatically and recorded in the hospital dashboard and the state health records system.

3.3 Ordering, verification and safeguards

- **Who can order:** only staff at facilities registered under a contract. **Patients cannot order**, and the drone never delivers to a home.
- **Prescription / requisition:** every order carries the treating doctor's requisition (blood requisition form, ASV indent). The doctor's login is tied to their National Medical Commission / State Medical Council registration number.
- **Fraud checks:** orders are compared with the facility's normal usage; unusual quantities are flagged and the hub calls the facility back before dispatch.
- **Locating the facility:** every hub and facility is registered with fixed GPS coordinates and a QR-coded drop zone. The dispatch engine picks the nearest hub that has the item in stock and a free drone.
- **Credible medicine:** stock comes only from government or licensed hospital blood banks and pharmacies, never from SkyPulse. Pods are sealed and tamper-evident, temperature-logged, and released only against an OTP.
- **Teleconsultation:** the app can connect the PHC doctor to a district-hospital specialist (similar to eSanjeevani) for advice such as ASV dosage. It does not prescribe to patients at home.

3.4 Technology used

- DGCA Digital Sky: drone registration (UIN), Remote Pilot Certificates, NPNT flight permission, airspace maps.
- Dispatch engine and multi-drop route optimiser.
- Machine-learning demand forecasting (for example, snakebite peaks in the monsoon) to pre-position stock.
- 4G telemetry streaming position, battery health and payload temperature every second.
- APIs to 108 emergency service, e-Raktkosh (national blood bank network) and hospital records.
- Encrypted links, role-based access, audit logs; data stored in India under the DPDP Act 2023.

3.5 Schedule, quality and KPIs

Hubs run two shifts, 6 am to 10 pm, 330 days a year, with a pilot on call at night for critical cases (subject to DGCA night-flight approval). Quality is managed through SOPs, digital pre- and post-flight checklists, preventive maintenance by flight hours, temperature-excursion alarms and quarterly safety audits.

KPI	Target
Emergency order-to-delivery time	< 35 minutes for 95% of orders
Flight success rate	> 99.5%
Cold-chain compliance	100% of pods within 2–8 °C
Fleet availability	> 90%
Customer satisfaction (NPS)	> 60

SCRIPT **Member 3 · Slide 6 · about 1 minute**

Our operations plan: from order to delivery in under 35 minutes.

It takes eight steps. A doctor orders. The hub pharmacist verifies the requisition in about three minutes. The medicine is packed in a sealed pod with a temperature logger. The pilot runs pre-flight checks, and our system gets airspace clearance on the government's Digital Sky platform. The drone flies for 15 to 25 minutes while we track it live. It drops the pod at the health centre, the staff confirm with an OTP, and the drone returns for a battery swap. Billing happens automatically.

Only registered hospitals can order, never the public. Every doctor's login is linked to their medical council registration, and unusual orders are flagged before we fly.

We measure ourselves on clear targets: under 35 minutes for 95 percent of emergencies, over 99.5 percent flight success, and every pod kept between 2 and 8 degrees.

4 Marketing plan

Slide 7 · Member 3

4.1 Target market and segmentation

Segment	Customers	What they need	How they buy
Public health (B2G), primary	District hospitals, CHCs, 2,138 PHCs, NHM Karnataka	Blood, ASV, oxytocin, vaccines, TB/HIV samples	Annual contracts via tenders / GeM
Private hospitals (B2B)	Hospital chains, nursing homes	Blood components, rare drugs, inter-branch samples	Monthly subscription
Diagnostics (B2B)	Labs and collection centres	Patient samples to central labs	Per delivery
Blood banks and NGOs	Red Cross and similar	Redistributing blood before expiry	Per delivery

Positioning statement: for hospitals and health departments that cannot afford to lose the golden hour, SkyPulse is the medical drone network that delivers blood and life-saving medicine in under 30 minutes with a verified cold chain, at a lower cost than road transport.

4.2 The 4 Ps

P	Strategy
Product	Four tiers on one platform: Emergency, LabLink (samples), Resupply (district contract), Priority (subscription SLA). A district can start with one route and grow.
Price	Emergency ₹1,400 → ₹1,550 by Y5 (value-based: below the ₹2,902 road cost). Scheduled ₹500 → ₹575 (cost-plus over ₹240 variable cost). Subscription ₹25,000/month. District contract ₹36 lakh/year.
Place	Government: NHM tenders and the GeM portal (DPIIT startups are exempt from prior turnover/experience criteria). Private: direct sales in Bengaluru Rural, Chikkaballapur, Kolar, Tumakuru. Partners: e-Raktkosh, Red Cross, 108 operator.
Promotion	Live demo flights for District Health Officers and Deputy Commissioners; talks at CME events and IMA meetings; case-study videos; press at milestones (first ASV delivery, 1,000th blood unit); Kannada/English onboarding kits for PHC staff.

4.3 Sales forecast and marketing budget

Driver	Y1	Y2	Y3	Y4	Y5
Deliveries	19,800	55,440	1,10,880	1,78,200	2,77,200
Partner hospitals (subscriptions)	8	20	40	65	95
District contracts	1	2	4	6	8
Revenue (₹ lakh)	221.4	573.9	1,153.8	1,899.0	2,977.7
Marketing budget (₹ lakh)	17.7	40.2	69.2	94.9	148.9

Controls: monthly review of customer acquisition cost vs contract value, deliveries per partner, NPS and repeat orders. Marketing spend falls from 8% of revenue in Year 1 to 5% by Year 4 as referrals and contracts take over.

SCRIPT **Member 3 · Slide 7 · about 1 minute**

Our marketing plan. Our main customer is the public health system: district hospitals and primary health centres, buying through the National Health Mission. That is where the need is greatest. Our second segment is private hospitals and labs: they pay faster, adopt sooner, and give us the success stories that win government contracts.

Product: four service tiers on one platform. Price: ₹1,400 per emergency delivery, which is less than half of what a road trip costs. Place: government tenders and the GeM portal, plus direct sales to hospital chains. Promotion: nothing convinces a health officer like a live demonstration flight, so that is where we start, backed by medical conferences and case-study videos.

With this plan, revenue grows from ₹2.2 crore in Year 1 to nearly ₹30 crore in Year 5.

5 Organizational plan

Slide 8 · Member 4

5.1 Form of ownership

Private Limited Company under the Companies Act, 2013, with **DPIIT startup recognition**. Reasons:

- Limited liability for the founders.
- Can raise equity from angel and venture investors and create an ESOP pool.
- DPIIT benefits: Startup India Seed Fund grant, 3-year income-tax holiday under Section 80-IAC (subject to approval), relaxed public procurement rules.
- Credibility with hospitals and government departments.

5.2 Management team

Role	Member	Responsibilities
Chief Executive Officer (CEO)	Member 1	Strategy, fundraising, government and hospital partnerships
Chief Technology Officer (CTO)	Member 2	Software platform, drone integration, IoT pod, cyber-security
Head of Flight Operations (COO)	Member 3	Hubs, pilots, DGCA compliance, safety, maintenance
Head of Finance & Partnerships (CFO)	Member 4	Accounts, pricing, tenders, billing, legal

5.3 Structure and authority

- **Board of Directors:** CEO, CTO and one seed-investor nominee; an independent director with healthcare experience joins in Year 2.
- **CEO** → CTO (software and IoT engineers), COO (hub managers → pilots and technicians; safety officer), CFO (sales, accounts, customer success).
- Spending above ₹10 lakh, new hubs and new shares need Board approval.
- The Head of Flight Operations can ground the whole fleet at any time for safety.

5.4 Advisory board

- A public-health physician or retired District Health Officer
- A DGCA-certified flight instructor / aviation safety expert
- A chartered accountant
- A BMSCE faculty mentor

5.5 Shareholding and staffing

Shareholder	At start	After Seed (₹5 Cr at ₹20 Cr pre-money)	After Pre-Series A (₹6 Cr at ₹40 Cr pre-money)
Founders (4, equal)	100%	72.0%	62.6%
ESOP pool	–	8.0%	7.0%

Shareholder	At start	After Seed (₹5 Cr at ₹20 Cr pre-money)	After Pre-Series A (₹6 Cr at ₹40 Cr pre-money)
Seed investors	–	20.0%	17.4%
Pre-Series A investors	–	–	13.0%

Staff	Y1	Y2	Y3	Y4	Y5
Hub staff (6 per hub)	12	24	42	60	84
Corporate team	14	16	19	21	24
Total employees	**26**	**40**	**61**	**81**	**108**

SCRIPT **Member 4 · Slide 8 · about 1 minute**

Our organizational plan. SkyPulse will be a Private Limited Company, recognised as a startup by DPIIT. This limits our personal liability, lets us raise investment, and gives us access to the Startup India Seed Fund and tax benefits.

The four of us split the leadership: [Member 1] as CEO handles strategy and partnerships, [Member 2] as CTO builds the software and the drone pod, [Member 3] runs flight operations and safety, and I handle finance and government tenders.

We report to a Board that includes an investor representative, and we are advised by a public-health doctor, an aviation safety expert and a chartered accountant. One rule is non-negotiable: our head of flight operations can ground the entire fleet at any time for safety.

We start with 26 people and grow to 108 by Year 5. After two funding rounds, the founders still own about 63 percent of the company.

6 Risk assessment

Slide 9 · Member 4

6.1 SWOT analysis

Strengths

- 15–30 min vs 1–4 hours by road
- Verified cold chain with temperature logs
- Cheaper per delivery than road
- Strong software and IoT skills in the team

Weaknesses

- Young team, no aviation track record
- 5 kg payload per flight
- Weather and approval dependent
- High upfront cost per hub

Opportunities

- BVLOS framework expected by end-2026
- NHM budgets for last-mile delivery
- Drone Shakti, PLI, DPIIT benefits
- Other states; organs and vaccine logistics

Threats

- Delays in BVLOS / night-flight approvals
- Well-funded competitors entering
- An accident damaging public trust
- Slow government payments

6.2 Risk register

Risk	Likelihood	Impact	Mitigation
BVLOS / corridor approval delay	Medium	High	Start on line-of-sight routes; join ICMR / state pilot corridors; compliance officer from day one
Drone crash / technical failure	Low	High	Type-certified drones, parachute, geofencing, redundant GPS, maintenance, full insurance
Monsoon grounds flights	High	Medium	Weather-aware scheduling; plan assumes only 330 flying days; road-courier fallback
Cold-chain breach	Low	High	Validated pods, live temperature alarms, quarantine SOP, product-loss insurance
Cyber-attack / data breach	Low	High	Encryption, India-hosted data, DPDP Act compliance, annual penetration test
Slow government payments	High	Medium	Mix of private subscriptions and govt contracts; milestone billing; cash buffer above ₹2 Cr
Competition from funded players	Medium	Medium	Lock in multi-year district contracts early; focus on rural emergency
Battery fire at hub	Low	High	Fire-rated battery room, charging protocols, training
Public concern (noise, privacy)	Low	Low	Community meetings; no camera recording on routes; flights at 120 m

6.3 Sensitivity analysis (re-run in the Excel model)

Scenario	Y5 revenue (₹ L)	Y3 PAT (₹ L)	Y5 PAT (₹ L)	Lowest cash (₹ L)	Enterprise value (₹ L)
Base case	2,977.7	140.9	695.4	248.3	1,227.7
Utilisation -20%	2,496.8	29.2	452.5	228.2	412.7
Prices -15%	2,617.0	17.8	438.4	226.1	359.4
Drone cost +30%	2,977.7	108.0	642.4	197.9	860.9
Govt contracts delayed 1 yr	2,905.7	75.6	644.1	215.2	991.0

In every scenario the company stays cash-positive and is profitable from Year 3. Lower utilisation or lower prices push cash payback beyond Year 5, so those are the risks we watch most closely.

6.4 Contingency plans

- **BVLOS delayed 12+ months:** continue line-of-sight routes with relay pilots and pause new hubs, extending cash runway by about 9 months.
- **Utilisation 20% below plan:** fill idle flight hours with private lab routes and delay drone purchases.
- **Serious incident:** ground the fleet immediately, run an independent investigation, notify DGCA, restart only after fixes.
- **Funding delayed:** apply for SIDBI / Karnataka Startup Cell debt and slow corporate hiring.

SCRIPT **Member 4 · Slide 9 · about 1 minute**

Every business has risks; here are ours and how we handle them.

The biggest is regulation: long-range flights need approvals that are still being finalised. So we start on shorter line-of-sight routes and join government pilot corridors. Weather is the most likely risk, since the monsoon grounds flights, so our plan assumes only 330 flying days and keeps a road courier as backup. A crash is unlikely but serious, so we use certified drones with parachutes, geofencing and full insurance. Government payments can be slow, so private subscriptions keep cash coming in, and we always hold over ₹2 crore in reserve.

We also stress-tested our financial model. With 20 percent fewer deliveries, 15 percent lower prices, 30 percent costlier drones, or government contracts a year late, SkyPulse still never runs out of cash and is profitable from Year 3.

7 Financial plan

Slide 10 · Member 4 · all figures in ₹ lakh unless stated; Year 1 = FY 2027-28

All estimates come from a **formula-driven Microsoft Excel model** (SkyPulse_Financial_Model.xlsx). Every number links to one Assumptions sheet, so changing any input recalculates the whole plan: revenue, costs, P&L, cash flow, balance sheet, break-even, NPV/IRR and sensitivity.

7.1 Key assumptions

Assumption	Value
Hubs (Y1 → Y5)	2, 4, 7, 10, 14 (6 drones each)
Deliveries per drone per day	5, 7, 8, 9, 10 (12 is physically possible)
Operating days per year	330
Emergency share of deliveries	35% → 30% (Rwanda: 40%)
Variable cost per delivery	₹240 (energy ₹12, battery wear ₹55, packaging ₹70, maintenance ₹85, telecom ₹18)
Drone / hub cost	₹14 lakh / ₹22 lakh
Depreciation	Straight line, 5 years
Tax / discount rate	25% (after loss set-off) / 18%

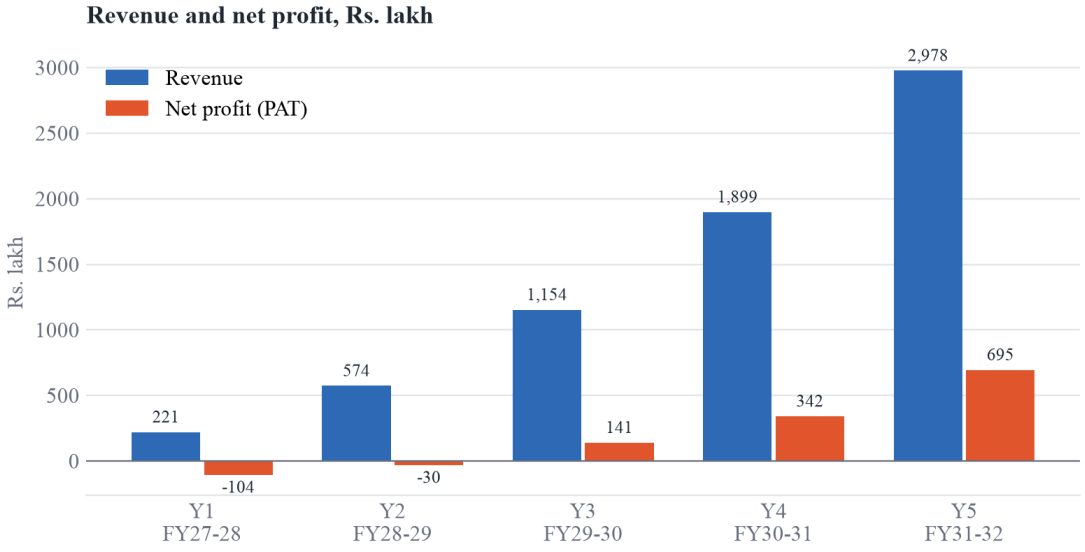
7.2 Funding: sources and applications (Years 1–2)

Sources	₹ lakh	Applications	₹ lakh
Founders' capital	50.0	Drones (fleet of 24)	336.0
Startup India Seed Fund grant	20.0	Hub setup (4 hubs)	88.0
Seed round (Y1)	500.0	Technology platform	75.0
Pre-Series A round (Y2)	600.0	Less: net operating cash generated (Y1 loss, Y2 profit)	(20.4)
		Reserve for Y3 expansion to 7 hubs	691.4
Total sources	**1,170.0**	**Total applications**	**1,170.0**

7.3 Projected income statement

₹ lakh	Y1	Y2	Y3	Y4	Y5
Total revenue	**221.4**	**573.9**	**1,153.8**	**1,899.0**	**2,977.7**
Less: variable costs	47.5	133.1	266.1	427.7	665.3
Gross profit	**173.9**	**440.8**	**887.7**	**1,471.3**	**2,312.4**
Less: fixed operating costs	223.5	370.8	578.0	780.9	1,062.5
EBITDA	**(49.7)**	**70.0**	**309.7**	**690.3**	**1,249.9**
EBITDA margin	-22.4%	12.2%	26.8%	36.4%	42.0%

₹ lakh	Y1	Y2	Y3	Y4	Y5
Less: depreciation	54.4	99.8	166.4	234.0	322.8
Profit before tax	(104.1)	(29.8)	143.3	456.3	927.1
Less: tax	0.0	0.0	2.4	114.1	231.8
Net profit (PAT)	**(104.1)**	**(29.8)**	**140.9**	**342.3**	**695.4**
Net margin	-47.0%	-5.2%	12.2%	18.0%	23.4%



Revenue and net profit by year

7.4 Cash flow and balance sheet

₹ lakh	Y1	Y2	Y3	Y4	Y5
Cash from operations	(49.7)	70.0	307.3	576.3	1,018.2
Capital expenditure	(272.0)	(227.0)	(333.0)	(338.0)	(444.0)
Funding received	570.0	600.0	0.0	0.0	0.0
Closing cash	**248.3**	**691.4**	**665.7**	**903.9**	**1,478.1**
Net fixed assets	217.6	344.8	511.4	615.4	736.6
Total assets	**465.9**	**1,036.2**	**1,177.1**	**1,519.3**	**2,214.7**
Share capital + grant	570.0	1,170.0	1,170.0	1,170.0	1,170.0
Retained earnings	(104.1)	(133.8)	7.1	349.3	1,044.7

The balance sheet balances in every year, and closing cash never falls below ₹248 lakh.

7.5 Break-even analysis

Break-even deliveries = (fixed costs + depreciation − subscription and contract revenue) ÷ contribution per delivery, where contribution = average delivery fee − ₹240 variable cost.

	Y1	Y2	Y3	Y4	Y5
Contribution per delivery (₹)	575	557	562	595	628
Break-even deliveries	37,897	60,785	85,410	1,01,504	1,29,448
Actual deliveries	19,800	55,440	1,10,880	1,78,200	2,77,200
Margin of safety	-91.4%	-9.6%	23.0%	43.0%	53.3%

SkyPulse is below break-even in Years 1–2 while the network is built, crosses it in **Year 3** (23.0% margin of safety) and reaches 53.3% by Year 5.

7.6 Investment appraisal

Measure	Value
Total funding required	**₹11.7 Cr**
NPV of 5-year free cash flows @ 18%	₹(27.1) lakh (network still being built)
Enterprise value incl. terminal value (5× Y5 FCF)	₹1,227.7 lakh
IRR incl. terminal value	74.1%
Cash payback	Y5 FY31-32
Cost per delivery Y1 → Y5	₹1,644 → ₹740 (road: ₹2,902)

SCRIPT **Member 4 · Slide 10 · about 1.5 minutes**

Finally, the numbers. We built our estimates in a formula-driven Excel model, so every figure traces back to a stated assumption and the whole plan updates if any input changes.

We need ₹11.7 crore over two years: ₹50 lakh from the founders, a ₹20 lakh Startup India grant, a ₹5 crore seed round in Year 1 and a ₹6 crore round in Year 2. About ₹5 crore goes into drones, hubs and software; the rest is a reserve for expanding to seven hubs.

Revenue grows from ₹2.2 crore in Year 1 to ₹29.8 crore in Year 5. We make a loss in the first two years while building the network, break even in Year 3 with a net profit of ₹140.9 lakh, and reach ₹7.0 crore profit in Year 5.

As we scale, the cost of one delivery falls from ₹1,644 to ₹740, against ₹2,902 for a road trip. Including terminal value, the business is worth about ₹12.3 crore with an IRR of 74 percent, well above our 18 percent hurdle rate.

If asked: Why is the 5-year NPV slightly negative? Because Years 1–3 are spent building hubs. The free cash flows turn strongly positive in Years 4–5, so the value shows up once the terminal value (the business continuing beyond Year 5) is included, at a conservative 5× Year-5 cash flow.

8 Conclusion

Slide 11 · all members

- **Saves the golden hour:** blood, ASV and drugs in 15–30 minutes instead of 1–4 hours.
- **Commercially viable:** ₹29.8 Cr revenue by Year 5, profitable from Year 3, cash-positive throughout.
- **Right time, low risk:** Drone Rules 2021, PLI, Drone Shakti; survives every stress test.
- **Scalable:** 2 → 14 Karnataka hubs, and a model other states can adopt.
- **The ask:** a ₹5 crore seed round to launch 2 hubs and 12 drones in Chikkaballapur and Kolar and prove the model.

SCRIPT **Member 1 · Slide 11 · about 30 seconds**

To conclude: SkyPulse saves the golden hour by getting blood and medicine to rural patients in minutes, not hours. It is commercially viable, profitable from Year 3, and it stays solvent under every risk we tested. Policy is on our side, and the model can scale from Karnataka to the rest of India.

We are asking for ₹5 crore to launch our first two hubs in Chikkaballapur and Kolar. Our vision is simple: no patient in Karnataka should die waiting for blood or medicine. Thank you. We are happy to take your questions.

9 Likely viva questions and answers

Q1. Do patients need a prescription before placing an order?

Patients do not order at all. SkyPulse is hospital-to-hospital: only staff at registered PHCs and hospitals can order, and each order carries the treating doctor's requisition (blood requisition form or ASV indent). The hub pharmacist verifies it before packing.

Q2. Can the app arrange an emergency call with a doctor to prescribe?

The patient is already with a doctor or nurse at the PHC, so no prescribing call is needed. The app can connect the PHC doctor to a district-hospital specialist for teleconsultation (like eSanjeevani), for example on ASV dosage. It does not prescribe to patients at home.

Q3. How does it find the nearest hospital?

Every hub and facility is registered with fixed GPS coordinates and a QR-coded drop zone. The ordering facility is identified by its login, and the dispatch engine picks the nearest hub that has the item in stock and a free drone.

Q4. How is the medicine credible?

It comes only from government or licensed hospital blood banks and pharmacies, never from SkyPulse's own stock. Pods are sealed, tamper-evident and temperature-logged at 2–8 °C, and released only against an OTP at the receiving facility.

Q5. How is this better than calling 108 (India's 911)?

108 moves patients; SkyPulse moves medicine, so we complement 108. A CAG audit found Karnataka's 108 met its 30-minute target in only 72% of cases, and even after the patient reaches a PHC, blood still has to come by road in 1–4 hours. A drone brings it in 15–30 minutes and can avoid a risky long transfer. Our app also integrates with 108 dispatch.

Q6. How do you know a prescription is not fake?

Only contracted facilities can order; each doctor's login is tied to their NMC / State Medical Council registration; orders are checked against the facility's normal usage and unusual ones trigger a call-back; every order is logged with an audit trail; and the drone only ever lands at a registered facility, so medicine cannot be diverted to a private address.

Q7. Is flying drones legal in India?

Yes. Drone Rules 2021 allow flights up to 400 ft in green zones (about 90% of airspace) without prior permission. We need a UIN for each drone, Remote Pilot Certificates for pilots, NPNT compliance and third-party insurance. Long-range BVLOS needs approval; the framework is expected by end-2026, and we start on line-of-sight routes until then.

Q8. What if a drone crashes with blood on board?

Each drone has an emergency parachute, geofencing and return-to-home. The pod is sealed and insured, and the hub immediately dispatches a replacement by drone or road. A serious incident triggers fleet grounding and a DGCA-notified investigation.

Q9. What about the monsoon?

Our plan assumes only 330 flying days. The drones fly in light rain and winds up to 35 km/h; outside those limits we use a road-courier fallback contract.

Q10. How did you estimate the financials?

With a formula-driven Excel model: every number links to stated assumptions sourced from published studies (Gupta et al. 2025, Lancet Global Health 2022), vendor prices and government data. It includes P&L, cash flow, balance sheet, break-even, NPV/IRR and sensitivity scenarios.

Q11. Why will government buy from a startup?

DPIIT-recognised startups are exempt from prior turnover and experience criteria in public procurement, and ICMR and states are already running drone pilots. We start with demo flights and private hospitals to build a track record.